Model Lineage — every training config, in training order, mapped to the research

The experiments that produced these models are in [[EXPERIMENT_CATALOG]], newest first.

> ⚠ **2026-08-18: every joint_ie / 137k number in this document predates a data repair and is superseded.** 45 corpora shipped overlapping train/val/test, and the scaling configs additionally paired regenerated train files against frozen val slices (252 train-in-val, 22 val-in-test). All were repaired, the slices rebuilt, and the four scaling points now gate CLEAN. The curve is being re-run from scratch — see `JOINT_IE_SCALING.md` and [[lambda-137k-curve-restart]]. Numbers below are kept as the record of what was measured and believed at the time; do not compare new results against them.

Status: index. Date: 2026-08-17; Stage 6 extended **2026-11-06** through **`eb22g`** (`PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`). `tools/train/config/` holds 251 YAML files (66 under `base/`); the stages below cover the ones that are links in a lineage.

Every model we trained is a chain: some base checkpoint, then one or more training stages, each stage a YAML in `tools/train/config/`. This document walks those stages in the order they run and, for each config, names the working paper that motivates or reports it. Read a row bottom-up and you get a model's provenance; read a stage top-down and you get every arm we ran at that stage.

Scope note: [[PIPELINES]] maps the *inference* path (text in, records out). This maps the *training* path. They do not overlap.

Terminology — "warm start" means three different things here

The configs use one phrase for three distinct operations. The stage numbering below keys off the mechanism, not the word:

usage	mechanism	where
cold start	`model.encoder:` — `from_encoder`, fresh GLiNER2 heads on a raw encoder	Stage 2
warm start (external)	`model.pretrained: fastino/...` — continues heads Fastino trained	Stage 4
warm start (intermediate stage)	`model.pretrained: ./out/...` — continues heads *we* trained, as a deliberate mid-training stage	Stage 3

`architecture: boundary` marks the boundary head; its absence means the older span head. The two are not interchangeable — warm-starting across them raises `ArchitectureMismatchError`.

The chains

```
A. Fastino span line (the workhorse)
   fastino/gliner2-{base,large,multi}-v1  --[Stage 4]-->  ~30 task fine-tunes
                                          --[Stage 5]-->  casualty-* (EKF extractor)

B. mmBERT span line (head-init question)
   jhu-clsp/mmBERT-{small,base}  --[Stage 1]-->  loss sweep
                                 --[Stage 2]-->  mmbert-base-rams  --[Stage 4]--> mmbert-base-wikievent
                                 --[Stage 2]-->  scaling-mmbert-{10k,40k,100k}

C. mmBERT boundary line (Paper 2 — the 137K scaling spine)
   jhu-clsp/mmBERT-base
      --[Stage 2]--> joint-boundary-mmbert-{10k,40k,100k,137k}
                     |                          |
                     |                          +--[Stage 3]--> warmstart-natural(-*), warmstart-anch
                     |                          |                joint-boundary-warmstart-struct
                     |                          |                    |
                     |                          |                    +--[Stage 4]--> rams-clean-{b,c}
                     |                          +--[Stage 4]--> rams-clean-a-base137k
                     +--[Stage 4]--> joint-boundary-{rams,redocred}-{10k,40k,100k,137k}
                                      |
                                      +--[Stage 4]--> {maven,casie}-tier2-{control,eventre

D. mmBERT event-record + junction + coref line (Paper 3 — the eb17..eb22g spine)
   jhu-clsp/mmBERT-base
      --[Stage 6]--> eb17-best --> eb18-balanced --> eb19 (junction) --> eb21 (negatives k=20, coref li
                                                                          |
                                                                          +--> corefC-*, rolesym-*
                                                                          |
                                                                          +--> eb22b --> eb22c (offset
                                                                                          |
                                                                                          +--> eb22d
                                                                                                 |
                                                                                                 +-->
```

Chain C is the 137K scaling spine (`JOINT_IE_SCALING.md`); Chain D is the document-level event and coreference spine (`PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`), culminating in `eb22g` .

Stage 0 — Bases we did not train

No config; these are the roots every chain starts from.

base	what it is	working paper
`fastino/gliner2-base-v1`	DeBERTa-v3-base, 254K examples, span + structure heads, **no event or relation head**	[[FASTINO_GLINER2_TRAINING]] §1, §3
`fastino/gliner2-large-v1`	the large counterpart	[[FASTINO_GLINER2_TRAINING]]
`fastino/gliner2-multi-v1`	mDeBERTa-v3 multilingual counterpart	[[FASTINO_GLINER2_TRAINING]]
`microsoft/deberta-v3-base`	raw encoder, 512 positions → 384-word window	[[PAPER_0_FOUNDATION]] §9.5
`jhu-clsp/mmBERT-base` / `-small`	raw encoder, 8192 positions — whole documents in one pass	[[PAPER_0_FOUNDATION]] §9.5

Why it matters downstream: base-v1 has no event head, so every event capability in Stage 4 is being learned from a head that was never trained on events. That is the head-init bottleneck the whole program circles ([[FASTINO_GLINER2_TRAINING]] §4-5).

Stage 1 — Struct-loss selection (cold start, small scale)

Which structure loss, decided before spending on scale. All `from_encoder` on mmBERT.

config	init	data	working paper	role
`mmbert-small.yaml`	mmBERT-small	ACE 2005 + stockmark_jpn, klue_re	[[PAPER_0_FOUNDATION]] §6	focal baseline
`mmbert-small-focal.yaml`	mmBERT-small	WikiEvents	[[PAPER_0_FOUNDATION]] §6, §10.3	focal arm
`mmbert-small-bce.yaml`	mmBERT-small	WikiEvents	[[PAPER_0_FOUNDATION]] §10.3	plain BCE arm
`mmbert-small-bce-posweight.yaml`	mmBERT-small	WikiEvents	[[PAPER_0_FOUNDATION]] §10.3	BCE + pos-weight — **the variant since adopted as the project default** (47 configs set it)
`mmbert-small-dice.yaml`	mmBERT-small	WikiEvents	[[PAPER_0_FOUNDATION]] §10.3	soft-Dice arm
`mmbert-small-bce-dice.yaml`	mmBERT-small	WikiEvents	[[PAPER_0_FOUNDATION]] §10.3	BCE + soft-Dice arm
`mmbert-small-asl.yaml`	mmBERT-small	WikiEvents	[[PAPER_0_FOUNDATION]] §10.3	asymmetric-loss arm
`mmbert-base.yaml`	mmBERT-base	25 corpora + 7 event corpora	[[PAPER_0_FOUNDATION]] §6, §7	scale-up of the chosen loss; the broadest span-arch mix we ran

`bce_posweight` with `pos_weight 8` is what the from-encoder configs downstream inherit. Note the provenance gap: adoption is documented, the *comparison* is not — [[PAPER_0_FOUNDATION]] §10.3's table is still `TBD` in every cell, so these seven arms are a sweep that was run and acted on but never written up. [[PROJECT_HISTORY]] is the implementation record for separating event loss out of `structure_loss`.

Stage 2 — Cold start (`from_encoder`, fresh heads)

The head-init question: can fresh heads be warmed from data alone, and how much data does it take? Two architecture arms.

2a. Span architecture

config	init	data	working paper	role
`mmbert-base-rams.yaml`	mmBERT-base	RAMS	[[PAPER_0_FOUNDATION]] §10.6, §9.5	the long-context from-encoder reference; pushed as `whr778/mmbert-base-rams`
`deberta-base-fromenc-rams.yaml`	deberta-v3-base	RAMS	[[PAPER_0_FOUNDATION]] §10.6	**encoder-isolation control** — same recipe, short context, separates encoder effect from head-init effect
`mmbert-base-combined.yaml`	mmBERT-base	synthetic + GLiNER multilingual + multi-task NER + RAMS (~96K, 2 epochs)	[[PAPER_0_FOUNDATION]] §10.7	the **treatment base** of the broad-data head-init A/B — **a negative result**; its WikiEvents fine-tune is the treatment arm in Stage 4b
`deberta-base-fromenc-synthetic.yaml`	deberta-v3-base	synthetic_sonnet5_1k	[[PAPER_0_FOUNDATION]] §7, §10.6	can synthetic data alone teach all five tasks from scratch
`scaling-mmbert-10k.yaml`	mmBERT-base	10 event corpora, 10K nested subsample	[[PAPER_0_FOUNDATION]] §10.7 §3-4, [[PAPER_0_FOUNDATION]] §10.7 §4	span data-scaling curve
`scaling-mmbert-40k.yaml`	mmBERT-base	same pool, 40K	[[PAPER_0_FOUNDATION]] §10.7	span data-scaling curve
`scaling-mmbert-100k.yaml`	mmBERT-base	same pool, ~100K (full)	[[PAPER_0_FOUNDATION]] §10.7	span data-scaling curve; arg F1 0.050 / 0.115 / 0.158 across the three points
`mmbert-base-masakhaner.yaml`	mmBERT-base	MasakhaNER 2.0 (20 African langs)	[[PAPER_0_FOUNDATION]] §7	multilingual from-encoder NER
`mmbert-base-masakhanews.yaml`	mmBERT-base	MasakhaNEWS (16 langs)	[[PAPER_0_FOUNDATION]] §7	multilingual from-encoder classification

config	init	data	working paper	role
`mmbert-base-wikiann.yaml`	mmBERT-base	WikiANN, **HF-streamed** (no disk)	[[PAPER_0_FOUNDATION]] §7	multilingual NER at 176-language scale; step-bounded because a stream has no length

The curve here is what [[PAPER_0_FOUNDATION]] §10.7 §2 predicted a bracket for and [[PAPER_0_FOUNDATION]] §10.8 reports.

2b. Boundary architecture — the joint_ie bases

config	init	data	working paper	role
`joint-boundary-mmbert-10k.yaml`	mmBERT-base	3 relation + 10 event corpora, 10K total	[[JOINT_IE_SCALING]] §3b, §4	boundary cold-start base
`joint-boundary-mmbert-40k.yaml`	mmBERT-base	same, 40K	[[JOINT_IE_SCALING]] §4	boundary cold-start base
`joint-boundary-mmbert-100k.yaml`	mmBERT-base	same, 100K	[[JOINT_IE_SCALING]] §4	boundary cold-start base
`joint-boundary-mmbert-137k.yaml`	mmBERT-base	same, 136,787 (full)	[[JOINT_IE_SCALING]] §4	**the base everything in Stages 3-4 hangs off**; pushed as `whr778/gliner2-joint-boundary-mmbert-137k`

Deliberately separate output paths from `scaling-mmbert-*`: these do not overwrite the span runs, so the two architectures stay comparable ([[JOINT_IE_SCALING]] §3c).

Stage 3 — Intermediate warm-start stage (boundary)

Between the 137K base and the downstream task: does a broad multi-task stage help? All init from `./out/joint-boundary-mmbert-137k/best`, all `architecture: boundary`.

config	data	working paper	role
`joint-boundary-warmstart-struct.yaml`	`warmstart_mix` + 12 event corpora (val/test only), 30% replay	[[JOINT_IE_SCALING]] §7	adds STRUCTURE + NER; exists because mmbert-137k cannot do `[C]` record extraction at all
`warmstart-natural.yaml`	`mix_natural`	[[JOINT_IE_SCALING]] §7	**record-mode A/B, arm A** — `natural`; the control recipe for all of Phase 3
`warmstart-anchorless.yaml`	`mix_anchorless`	[[JOINT_IE_SCALING]] §7	**arm B** — `anchorless`; byte-identical mixture bar `record_metadata`. Learns nothing
`warmstart-natural-seed43.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §4	**noise floor** — control recipe at a second seed; any |delta| below this gap is unreadable
`warmstart-natural-evw05.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §4	run 1 dose-response, `task_loss_weights` events=0.5
`warmstart-natural-evw20.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §4	run 1, events=2.0
`warmstart-natural-evw40.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §4	run 1, events=4.0
`warmstart-natural-evwide2.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §10	run 2 extended-reach flat weight, w=2.0 → 12.5% of gradient
`warmstart-natural-evwide4.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §10	run 2, w=4.0 → 22.2% of gradient
`warmstart-natural-evpw08.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §10	run 2 per-task **pos_weight** 8.0 → 12.2% of gradient
`warmstart-natural-evpw16.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §10	run 2, pos_weight 16.0 → 17.8%
`warmstart-natural-evpw32.yaml`	`mix_natural`	[[PROJECT_HISTORY.md]] §10	run 2, pos_weight 32.0 → 27.1%

config	data	working paper	role
`warmstart-natural-gist.yaml`	`mix_natural`	[[EKF_MHT_DESIGN]] §27.4-27.9	GIST guide-embedding arm — the query-axis hard-negative test

The `evw*` / `evpw*` arms differ from `warmstart-natural.yaml` in exactly one field group (the loss weights) plus `output_dir` and `experiment_name`. They are generated, not hand-edited. Run 1's flat weight was a **null lever**; §10 explains why and computes run 2's doses rather than guessing them.

Stage 4 — Task training (the downstream fine-tune)

4a. On the boundary bases — Paper 2's curve and its Tier 2 follow-ups

config	init	data	working paper	role
`joint-boundary-rams.yaml`	137k base	RAMS	[[JOINT_IE_SCALING]] §4	**template** — the per-base copies below are generated from it
`joint-boundary-rams-10k.yaml`	10k base	RAMS	[[JOINT_IE_SCALING]] §4	event downstream, curve point
`joint-boundary-rams-40k.yaml`	40k base	RAMS	[[JOINT_IE_SCALING]] §4	event downstream, curve point
`joint-boundary-rams-100k.yaml`	100k base	RAMS	[[JOINT_IE_SCALING]] §4	event downstream, curve point
`joint-boundary-rams-137k.yaml`	137k base	RAMS	[[JOINT_IE_SCALING]] §4	event downstream; pushed as `whr778/gliner2-joint-boundary-rams-137k`
`joint-boundary-redocred.yaml`	137k base	Re-DocRED	[[JOINT_IE_SCALING]] §4	**template** for the relation downstream
`joint-boundary-redocred-10k.yaml`	10k base	Re-DocRED	[[JOINT_IE_SCALING]] §4	relation downstream, curve point
`joint-boundary-redocred-40k.yaml`	40k base	Re-DocRED	[[JOINT_IE_SCALING]] §4	relation downstream, curve point
`joint-boundary-redocred-100k.yaml`	100k base	Re-DocRED	[[JOINT_IE_SCALING]] §4	relation downstream, curve point
`joint-boundary-redocred-137k.yaml`	137k base	Re-DocRED	[[JOINT_IE_SCALING]] §4	relation downstream, curve point
`rams-clean-a-base137k.yaml`	137k base	RAMS	[[JOINT_IE_SCALING]] §4	**CONTROL** — the published 137K recipe re-run on current code, after ~10 commits touched the loss path
`rams-clean-b-warmstart.yaml`	`whr778/gliner2-warmstart-natural-clean`	RAMS	[[JOINT_IE_SCALING]] §4, [[PROJECT_HISTORY.md]]	**TREATMENT** — does routing through the Stage 3 `mix_natural`

config	init	data	working paper	role
				stage help the event downstream
`rams-clean-c-evwide2.yaml`	`whr778/gliner2-warmstart-natural-evwide2-clean`	RAMS	[[PROJECT_HISTORY.md]] §10	as B, but the Stage 3 arm carried the event-weighted loss. Read against B, not A
`maven-tier2-control.yaml`	`whr778/gliner2-joint-boundary-rams-137k`	MAVEN	[[JOINT_IE_SCALING]] Tier 2	does the RECORD head recover instances the mention path cannot express — control arm
`maven-tier2-eventrecords.yaml`	same	MAVEN	[[JOINT_IE_SCALING]] Tier 2	treatment arm. MAVEN over CASIE: 12x the instance supervision
`casie-tier2-control.yaml`	same	CASIE	[[JOINT_IE_SCALING]] Tier 2	control arm on the multi-instance-dense corpus (94.5% of docs repeat an event type)
`casie-tier2-eventrecords.yaml`	same	CASIE	[[JOINT_IE_SCALING]] Tier 2	treatment arm

The 12 curve arms are Stage 2 × Stage 4 (4 bases × {RAMS, Re-DocRED} plus the base's own metrics). All must be read at a **matched threshold** — [[JOINT_IE_SCALING]] §4b records the defect that hid this once already.

4b. On the Fastino span bases — English events

config	init	data	working paper	role
`gliner2-base-v1-rams.yaml`	base-v1	RAMS	[[PAPER_0_FOUNDATION]] §10.4	warm-start event reference; the head-init contrast to `mmbert-base-rams`
`gliner2-base-v1-casie.yaml`	base-v1	CASIE	[[PAPER_0_FOUNDATION]] §10.4	event sweep
`gliner2-base-v1-docee.yaml`	base-v1	DocEE	[[PAPER_0_FOUNDATION]] §10.4	event sweep
`gliner2-base-v1-wikievents.yaml`	base-v1	WikiEvents	[[PAPER_0_FOUNDATION]] §10.1	the blind-test headline run
`gliner2-base-v1-events-english.yaml`	base-v1	CASIE + DocEE + MAVEN + RAMS + WikiEvents	[[PAPER_0_FOUNDATION]] §10.4	joint English-event training
`gliner2-large-v1-rams.yaml`	large-v1	RAMS	[[PAPER_0_FOUNDATION]] §10.4	large-model sweep
`gliner2-large-v1-casie.yaml`	large-v1	CASIE	[[PAPER_0_FOUNDATION]] §10.4	large-model sweep
`gliner2-large-v1-docee.yaml`	large-v1	DocEE	[[PAPER_0_FOUNDATION]] §10.4	large-model sweep
`gliner2-large-v1-maven.yaml`	large-v1	MAVEN (event form)	[[PAPER_0_FOUNDATION]] §10.4	large-model sweep
`gliner2-large-v1-wikievents.yaml`	large-v1	WikiEvents	[[PAPER_0_FOUNDATION]] §10.4	large-model sweep. **Header comment says multi-v1; the `pretrained:` field says large-v1 — trust the field**
`gliner2-large-v1-events-english.yaml`	large-v1	5 English event corpora	[[PAPER_0_FOUNDATION]] §10.4	joint English-event training, large
`gliner2-multi-wikievents.yaml`	multi-v1	WikiEvents	[[PAPER_0_FOUNDATION]] §10.1	SOTA-comparison probe on WikiEvents

config	init	data	working paper	role
`mmbert-base-wikievents.yaml`	`whr778/mmbert-base-rams`	WikiEvents	[[PAPER_0_FOUNDATION]] §10.6-10.7	**the one span-line two-stage chain**: 206 train docs is too few for fresh heads, so it inherits argument competence from the RAMS mmBERT. **This file served both §10.7 arms** — as checked in it is the *control* (RAMS base); the *treatment* run re-pointed `pretrained:` at `whr778/mmbert-base-combined`. The treatment's parent is not recoverable from the file, only from §10.7

4c. On the Fastino span bases — Chinese and multilingual

config	init	data	working paper	role
`gliner2-multi-v1-cmnee.yaml`	multi-v1	CMNEE (Chinese military news)	[[PAPER_0_FOUNDATION]] §10.4	Chinese doc-level events
`gliner2-multi-v1-duee.yaml`	multi-v1	DuEE 1.0	[[PAPER_0_FOUNDATION]] §10.4	full trigger + typed-argument Chinese events
`gliner2-multi-v1-chfinann.yaml`	multi-v1	ChFinAnn	[[PAPER_0_FOUNDATION]] §10.4	**trigger-free** — converter emits role-typed entities + multi-label classification, so the metric is classification F1
`gliner2-multi-v1-docfee.yaml`	multi-v1	DocFEE	[[PAPER_0_FOUNDATION]] §10.4	trigger-free and offset-free; same shape as ChFinAnn
`gliner2-multi-v1-events-chinese.yaml`	multi-v1	CMNEE	[[PAPER_0_FOUNDATION]] §10.4	joint Chinese-event training
`gliner2-multi-v1-events-all.yaml`	multi-v1	6 corpora, English + Chinese	[[PAPER_0_FOUNDATION]] §10.4	the widest event mix on the span architecture
`gliner2-multi-v1-masakhaner.yaml`	multi-v1	MasakhaNER 2.0	[[PAPER_0_FOUNDATION]] §7	warm-start counterpart to `mmbert-base-masakhaner`
`gliner2-multi-v1-masakhanews.yaml`	multi-v1	MasakhaNEWS	[[PAPER_0_FOUNDATION]] §7	warm-start counterpart to `mmbert-base-masakhanews`
`gliner2-multi-v1-wikiann.yaml`	multi-v1	WikiANN, HF-streamed	[[PAPER_0_FOUNDATION]] §7	warm-start counterpart to `mmbert-base-wikiann`

4d. On the Fastino span bases — relations, NER, synthetic

config	init	data	working paper	role
`gliner2-base-v1-redocred.yaml`	base-v1	RE-DocRED	[[PAPER_0_FOUNDATION]] §10.5, [[RE_DIFFERENCES]]	relation extraction; selected on relation F1, not loss
`gliner2-large-v1-redocred.yaml`	large-v1	RE-DocRED	[[PAPER_0_FOUNDATION]] §10.5	relation extraction, large
`gliner2-large-v1-docred.yaml`	large-v1	DocRED (original annotation)	[[PAPER_0_FOUNDATION]] §10.5	the un-re-annotated contrast to RE-DocRED
`gliner2-base-v1-biomed-ner.yaml`	base-v1	13 MTL-Bioinformatics-2016 corpora	[[PAPER_0_FOUNDATION]] §7	NER is what this head was trained for, so warm start is the strongest default
`gliner2-base-v1-maven.yaml`	base-v1	`data/maven_ner`	[[PAPER_0_FOUNDATION]] §10.4	MAVEN as **NER**, not as events — unlike `gliner2-large-v1-maven`. Only a train split is on disk
`gliner2-base-v1-mendeley-ed.yaml`	base-v1	Mendeley Event Detection	[[PAPER_0_FOUNDATION]] §10.4	one generic event type, no arguments; metric is trigger F1
`gliner2-base-v1-synthetic.yaml`	base-v1	synthetic_sonnet5_1k	[[PAPER_0_FOUNDATION]] §7	can our synthetic data teach all five tasks — **in-distribution gain only**; see the eval caveat below
`gliner2-base-v1-synthetic-haiku5k.yaml`	base-v1	synthetic_haiku45_5k	[[PAPER_0_FOUNDATION]] §7	the 5K Haiku-generated counterpart
`rams-baseword.yaml`	base-v1	RAMS, lemma/base-word form	[[TODO]]	lemmatized-argument arm;

config	init	data	working paper	role
				provisional — unswept threshold
`rams-duplicate-control.yaml`	base-v1	RAMS, `train.duplicate_control`	[[TODO]]	the matched control for `rams-baseword`

Stage 5 — Application training (the EKF extractor)

Downstream of Paper 0's models, upstream of Paper 1's tracker. Most `casualty-*` arms init from `fastino/gliner2-base-v1`; `ekf-frontend-mmbert` is a deliberate cold start; `casualty-full-multilingual` is the odd one out, warm-started from the boundary line instead (see its own row).

config	data	working paper	role
`ekf-frontend-mmbert.yaml`	189,284 records, English trigger→argument 798 → ~39,800	[[EKF_MHT_DESIGN]] §7.5-7.6	**cold start from `jhu-clsp/mmBERT-base`**, built to BE the pipeline's first stage. Beats `137k-clean` on all 8 held-out heads AND on wire-copy binding precision (67-100% vs 0-7.7%). **The best extractor in this line on real news.** It fails the pre-registered form gate, which counts firings rather than correct ones -- see §7.6. Still blocked for the router by the one-instance-per-type decode, not by data
`casualty-full-multilingual.yaml`	`casualty_ml`: real news, downsampled to the smallest of the three languages (20,901 each)	[[GATES]]	**warm-started from `gliner2-joint-boundary-mmbert-137k-v2-eb16`**, not base-v1. Pushed as `gliner2-casualty-multilingual`, now the pipeline's `--casualty-model` default (`3e2b357`). Beats the old English-only default on real Turkish and Chinese held-out news (F1 0.0847→0.2919 tr, 0.1900→0.3592 zh) and by 92x pooled RMSE on real Helene copy (3634.7 → 39.5), catching a 19,180-vs-a-true-49 misbinding in the old model along the way
`casualty-finetune.yaml`	`casualty_ft` (Sonnet-5 realized)	[[EKF_MHT_DESIGN]] §19-20	closes the extraction gap the §19 `missing`-role probe predicted: zero-shot precision 0.63 + confidence-cut selection bias
`casualty-multievent.yaml`	`casualty_multi`	[[EKF_MHT_DESIGN]] §20, [[COUNTING_LAYER]]	the §20 corpus had exactly one `casualty_report` in all 31,539 docs, so the count head only ever saw "1". This fixes that
`casualty-docee.yaml`	`casualty_docee`	[[EKF_MHT_DESIGN]] §20-21	successor to `casualty-multievent`: synthetic trajectories paired with **real** DocEE contexts

Stage 6 — The eb bases, 2026-09-22 onward (boundary, from `jhu-clsp/mmBERT-`

`base`)

base	what it was	status
eb17-best	every measured lesson to its date; first base with `validate_data` on (the record-anchor crash, 2026-09-22)	superseded
eb18-balanced	English event supervision, reach, capacity, global_decode; best epoch 4 of 5	provisional: under-trained (the LR ran out with the metric still rising)
eb19	the junction, hard role negatives, identity proposal gold; hand-stopped at epoch 7; warm-start base of the 2026-10 menu A/Bs	**judged broken** (2026-10-14); not a baseline to beat
eb20	—	**skipped** after a critical discovery
eb21	from scratch: label negatives at dose 20 (MENU_SPEC s8-9), unified-full-v3 labels (Title_Snake), the news55 app menu for eval, the entity-shuffle fix (be581a1), on the fastino v2.1.0 merge (`merge/upstream-20261014`; contained-span dedupe on)	**the base going forward**: the most likely usable events architecture
corefC-control / -treatment / -treatment-free	warm eb21, 2 epochs: 34% exact replay + the bought LLM news coref; entity-coref link weight 0 / 0.3 / 0.3 + GUM/LitBank/GAP ([[COREF_DATA_SOURCES]])	**DONE 2026-10-16**: treatment link learns news pronouns (AUC 0.811 vs 0.685); eb21 heads inside the floor; GUM/LitBank/GAP dropped
rolesym-control / -treatment	warm eb21, 2 epochs: exact eb21 replay without casie/rams_merged/wikievents + those three in full; ONLY `negative_gold_full_roles` differs ([[MENU_SPEC]] s10, [[LABEL_NEGATIVES_PLAN]] s5g)	**DONE 2026-10-18, POSITIVE**: news55 event_type F1 0.153 -> 0.541, poison gone on CASIE (2 -> 72 hits), heads inside the floor
eb22 (option A)	warm eb21 = rolesym-treatment + corefC-treatment	**CRASHED** at step 5,779 (coref clusters matched by text; fixed 590d038); not restarted
eb22b	eb22 as OPTION B, from scratch, with fastino's structural relation/span flags	smoke DONE + CLEAN 2026-10-19; memory 96.7% of 40 GB
eb22c	eb22b + OFFSET-anchored event gold (casie_off, cmnee_ner_off, wikievents_off), unified-full-v4	launched 2026-10-19/20; events swing at the score LEVEL after the LR peak ([[EB22C_SCORE_LEVEL_AUTOPSY]]); the collapse-trace control to 30k is the reference every arm is read against
eb22c-*-trace arms	one change each on `base/eb22c-ddp-trace.yaml` (bce marginal loss): seed43, sep,	corefdist QUALIFIES; geomgelu / clsfix NULL; relfix NEGATIVE; the rest inside the floor ([[EXPERIMENT_CATALOG]] 2026-10-20..24)

base	what it was	status
	encfix, colfix, geomgelu, tasklr4e5, corefdist (+ resume to epoch 2), relfix, clsfix	
eb22d	eb22c + tier-1 fixes (seed, DDP gradient averaging, by-name optimizer restore, epoch checkpoints, metric_sweep readout) + `coref_link_distance: mlp` 64 + `boundary_marginal_loss: bce`; `make_eb22d_config.py`	**TRAINED** (`whr778/gliner2-eb22d-trace-epoch`, epochs 1-2 on Hub; epoch 1 argument strict micro F1 0.1123, trigger 0.1608 at 0.1; full-val at 0.3 epoch 1 -> 2: trigger 0.1334 -> 0.3645, argument 0.0442 -> 0.0627)
`life-mdeberta-wikievents`	`microsoft/mdeberta-v3-base` (`512` window) with the `eb22d` boundary/record head on `wikievents_off`, 30 epochs (`trace/life-mdeberta-wikievents.yaml`)	**DONE 2026-10-28** (`whr778/gliner2-life-mdeberta-wikievents`): val @0.1 strict micro F1 entity 0.3746, event_type 0.1825, trigger 0.1487, argument 0.0107
eb22e	eb22d + `span_residual_zero_init: true` + `enable_rotary_endpoints: false` (`make_eb22e_config.py`)	**DONE (epoch 1)** on 1x H100 (`whr778/gliner2-eb22e-epoch`): entity 0.4262 (+0.064 vs eb22d), structure 0.2231 (+0.054), trigger 0.1608, argument 0.0755 at threshold 0.1. Paired role-candidate probe (`probe_role_candidates.py`): 97.1% of gold role fillers are candidates under scheduled injection 0.9559 (median assignment P 0.0155), dropping to 38.3% with injection off
eb22f	eb22e + split role-loss mean (`_list_field_bce`), withhold unresolved gold roles, mask overlapping partial-match candidates, `record_drop_anchor_field_loss: true`, assignment-score format gate (`make_eb22f_config.py`)	**DONE (epoch 1)** on 4x H100 (`whr778/gliner2-eb22f-epoch`), preceded by A10 64-doc overfit (`overfit-events-eb22e`, argument F1 @0.5 +0.055 on CASIE, +0.266 on CMNEE; median gold role P 0.032 -> 0.554 CASIE, 0.508 -> 0.920 CMNEE). Full mix epoch 1: entity 0.4679 (+0.042 vs eb22e), relation 0.1665 (+0.034), trigger 0.1619, argument 0.0767 at 0.1 (confirmed proposal reachability was the remaining bottleneck)
eb22g	eb22f + `pair_split_reduction: true`, `hard_negative_keep_all_when_absent: false`, `remove_events_prob: 0.0`, occurrence/coref target grouping in `RecordFieldTarget.values` (`make_eb22g_config.py`, `PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`)	**5 EPOCHS COMPLETE (`bcfe500c`, `checkpoint-epoch-1..5` with `training_state.pt` verified on `whr778/gliner2-eb22g`; blind test running)**: preceded by A100 64-doc overfit (`overfit-events-eb22g`, lifting @0.5 strict micro F1 to `0.5210` trigger / `0.4319` argument on CASIE and `0.9394` trigger / `0.8005` argument on CMNEE). Full-mix Epoch 5 val @0.5 strict micro F1: `event_argument` **0.3038** (`0.3046` Ep 4; relaxed **0.5342**, fair **0.5221**; 4.85x `eb22d` Ep

base	what it was	status
		2), `event_trigger` **0.5720** (R=0.7399, fair **0.5980**), `event` **0.6314** (relaxed **0.6858**), `event_cluster` **0.5735**, `event_cluster_argument` **0.2614** (`0.2655` Ep 4), `entity` **0.3703** (relaxed **0.4466**), `relation` **0.2443** (relaxed **0.3130**), `structure` **0.2712**, `classification` **0.7210**, `overall` **0.4178**, `head_min` **0.2443**. **New base for downstream EKF models**

Entity and classification scores do not compare across eb21 and earlier bases. eb21's training data differs by the entity position fix, the classification gold-as-negative fix (22.6% of views) and the contained-span dedupe (9.8% of entity matches); see `EXPERIMENT_CATALOG.md`, 2026-10-08 and 2026-10-14.

Appendix — configs that are not links in any lineage

config	working paper	what it is
`eval-preservation-ner.yaml`	[[PAPER_0_FOUNDATION]] §8	**eval-only.** Scores a checkpoint on general-domain NER base-v1 is already good at. Exists because the synthetic configs cannot answer their own question — their val measures in-distribution gain, not preservation. Run twice, same command, different `--checkpoint`
`parity-deberta-rams.yaml`	[[JOINT_IE_SCALING]], [[BOUNDARY_ARCHITECTURE]]	port validation. One short run exercises from_encoder + bce_posweight + metric sweep + global decode + sliding window + events on a real encoder. Not a result
`probe-nan-a.yaml`	[[BOUNDARY_ARCHITECTURE]] §13	NaN isolation, arm A: sdpa + bf16 (the failing baseline)
`probe-nan-b.yaml`	[[BOUNDARY_ARCHITECTURE]] §13	arm B: sdpa + fp32 (isolates dtype)
`probe-nan-c.yaml`	[[BOUNDARY_ARCHITECTURE]] §13	arm C: flash_attention_2 + bf16 (the fix)
`stopwords.yaml`	—	**not a training config.** A stopword supplement keyed by ISO 639-2 codes, merged with the `stopwordsiso` package by `build_stopwords()`

The three `probe-nan-*` configs are generated by `tools/train/make_nan_probe_configs.py` — do not hand-edit.

Reading the lineage backwards

Given a checkpoint, walk up:

pushed checkpoint	produced by	its parent
`whr778/gliner2-joint-boundary-mmbert-137k`	`joint-boundary-mmbert-137k.yaml`	mmBERT-base (raw)
`whr778/gliner2-joint-boundary-rams-137k`	`joint-boundary-rams-137k.yaml`	the 137k base
`whr778/gliner2-joint-boundary-warmstart-natural`	`warmstart-natural.yaml`	the 137k base
`whr778/gliner2-warmstart-natural-clean`	`warmstart-natural.yaml`, clean-data re-run *(inferred)*	the 137k base
`whr778/gliner2-warmstart-natural-evwide2-clean`	`warmstart-natural-evwide2.yaml`, clean-data re-run *(inferred)*	the 137k base
`whr778/mmbert-base-rams`	`mmbert-base-rams.yaml`	mmBERT-base (raw)
`whr778/mmbert-base-combined`	`mmbert-base-combined.yaml`	mmBERT-base (raw)
`whr778/mmbert-base-combined-wikievents`	`mmbert-base-wikievents.yaml`, **re-pointed** at the combined base	`whr778/mmbert-base-combined`
`whr778/scaling-mmbert-{10k,40k,100k}`	`scaling-mmbert-*.yaml`	mmBERT-base (raw)
`whr778/gliner2-base-v1-synthetic`	`gliner2-base-v1-synthetic.yaml`	`fastino/gliner2-base-v1`
`whr778/deberta-base-fromenc-synthetic`	`deberta-base-fromenc-synthetic.yaml`	deberta-v3-base (raw)
`whr778/gliner2-large-v1-docee`	`gliner2-large-v1-docee.yaml`	`fastino/gliner2-large-v1`
`whr778/gliner2-maven-tier2-*`	`maven-tier2-*.yaml`	`whr778/gliner2-joint-boundary-rams-137k`
`whr778/gliner2-eb21`	`base/eb21.yaml`	`jhu-clsp/mmBERT-base` (raw)
`whr778/gliner2-eb22d-trace-epoch`	`base/eb22d-trace.yaml` + `base/eb22d-trace-h100.yaml`	`jhu-clsp/mmBERT-base` (raw, epochs 1-2)
`whr778/gliner2-life-mdeberta-wikievents`	`trace/life-mdeberta-wikievents.yaml`	`microsoft/mdeberta-v3-base` (raw, 30 epochs)
`whr778/gliner2-eb22e-epoch`	`base/eb22e.yaml`	`jhu-clsp/mmBERT-base` (raw, epoch 1)
`whr778/gliner2-eb22f-epoch`	`base/eb22f.yaml`	`jhu-clsp/mmBERT-base` (raw, epoch 1, 4x H100)
`whr778/gliner2-eb22g`	`base/eb22g.yaml`	`jhu-clsp/mmBERT-base` (raw, 5 epochs)

A config is not a permanent record of the run that used it. `pretrained:` gets re-pointed between arms and the previous value leaves no trace in git if the edit was never committed — `mmbert-base-wikievents.yaml` above is the proven case, where only [[PAPER_0_FOUNDATION]] §10.7 records that a second arm existed and what its base was. Two rows marked *(inferred)* rest on checkpoint naming alone: the `-clean` pair is documented in no working paper, only in the `pretrained:` fields of `rams-clean-b` / `-c`. Treat both as claims to confirm against a run log, not as established lineage.

Two standing cautions

Contamination. Split uniqueness is a gate, not a nicety: any arm read across a train/val/test overlap is invalid regardless of how clean the lineage looks. See [[TODO]] for the known table.

Matched thresholds. Curve points and A/B arms must be compared at the same decision threshold. [[JOINT_IE_SCALING]] §4b documents a case where an unmatched threshold produced a result that survived review and was later retracted.

Matched regimes, and a matched instrument. Strict, relaxed and fair are three different measurements; [[EKF_MHT_DESIGN]] §7.6 records a gate written to compare a candidate's STRICT score against the incumbent's RELAXED one, which turned a doubling into an apparent halving. The same section records a threshold sweep that was inert for five measurements. When comparing two checkpoints, score BOTH with one command on one machine — the historical number in a table is not a substitute, and re-measuring the incumbent costs one extra eval.

Related

- [[RESEARCH_PROGRAM]] — which working paper feeds which of the four papers
- [[PAPER_3_DOCUMENT_EVENTS_AND_COREF]] — Paper 3, synthesizing the `eb17` → `eb22g` event and coreference line
- [[PIPELINES]] — the inference path these models are deployed into
- [[BOUNDARY_ARCHITECTURE]] — what the boundary head actually does with these configs
- [[PROJECT_HISTORY]] — chronological record, including the decisions later overturned